Practical :2

Q. Design Employee Payroll System using Inheritance, Interface & Polymorphism

Source: program.cs

Code :

```
using System;

namespace EmployeePayrollSystem
{
  // Interface
  interface IPayroll
  {
    void CalculateSalary();
  }

  // Base Class
  class Employee
  {
    protected int empId;
    protected string empName;

    public Employee(int id, string name)
    {
      empId = id;
      empName = name;
    }

    public virtual void DisplayDetails()
    {
      Console.WriteLine("Employee ID   : " + empId);
      Console.WriteLine("Employee Name : " + empName);
    }
  }

  // Derived Class - Full Time Employee
  class FullTimeEmployee : Employee, IPayroll
  {
    private double monthlySalary;

    public FullTimeEmployee(int id, string name, double salary)
```

```
    : base(id, name)
  {
    monthlySalary = salary;
  }

  public void CalculateSalary()
  {
    Console.WriteLine("Monthly Salary : ₹" + monthlySalary);
  }

  public override void DisplayDetails()
  {
    Console.WriteLine("\n----- Full Time Employee -----");
    base.DisplayDetails();
    CalculateSalary();
  }
}

// Derived Class - Part Time Employee
class PartTimeEmployee : Employee, IPayroll
{
  private int hoursWorked;
  private double hourlyRate;

  public PartTimeEmployee(int id, string name, int hours, double rate)
    : base(id, name)
  {
    hoursWorked = hours;
    hourlyRate = rate;
  }

  public void CalculateSalary()
  {
    double salary = hoursWorked * hourlyRate;
    Console.WriteLine("Salary : ₹" + salary);
  }

  public override void DisplayDetails()
  {
    Console.WriteLine("\n----- Part Time Employee -----");
    base.DisplayDetails();
    Console.WriteLine("Hours Worked : " + hoursWorked);
    Console.WriteLine("Hourly Rate  : ₹" + hourlyRate);
    CalculateSalary();
  }
```

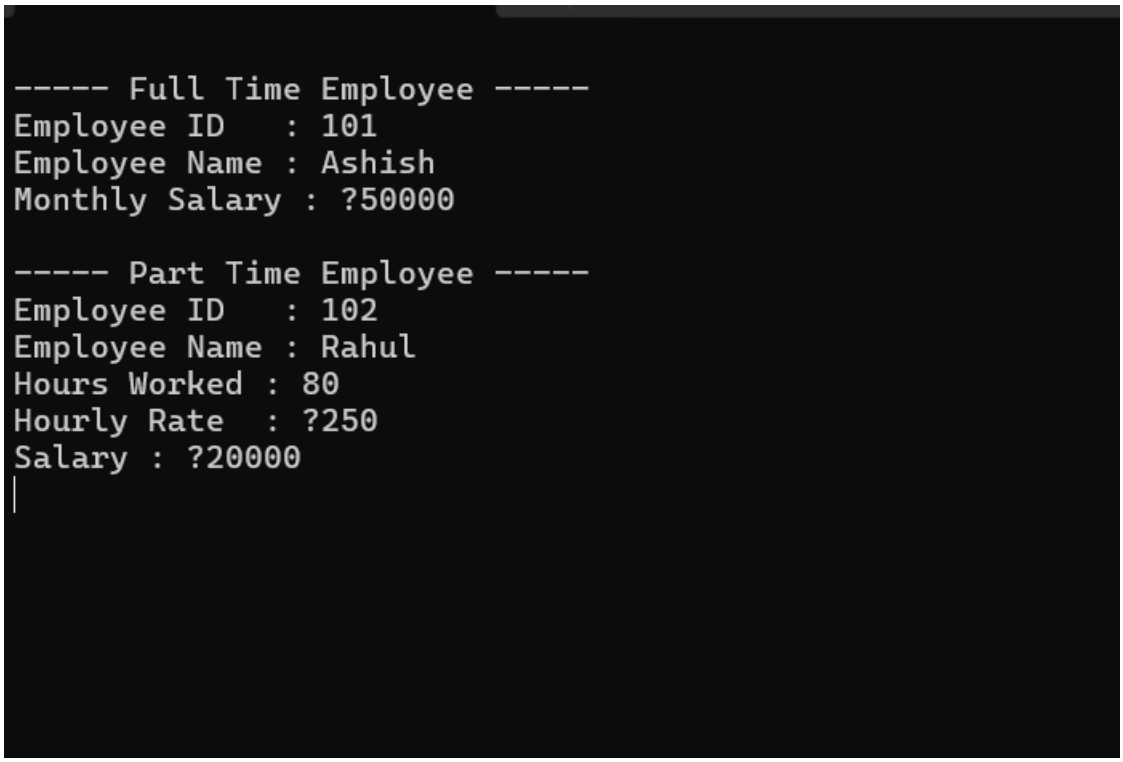
```
}

// Main Class
class Program
{
   static void Main(string[] args)
   {
      // Polymorphism
      Employee emp1 = new FullTimeEmployee(101, "Ashish", 50000);
      Employee emp2 = new PartTimeEmployee(102, "Rahul", 80, 250);

      emp1.DisplayDetails();
      emp2.DisplayDetails();

      Console.ReadLine();
```

Output: Screenshot:



GitHub URL :

https://github.com/Ashish-Agrawal-7322/.Net.git